

The Polymerase Chain Reaction

1. **(1 points) What is the PCR?**
 - a. PCR is a polymerase chain reaction. It amplifies and replicates a specific sequence of DNA in a test tube.
2. **(3 points) What is the purpose of the three different cycling temperatures?**
 - a. The three different cycling temperatures help control the stages of DNA synthesis. The first cycle, at 95° C, unwinds the DNA; the second at 60° C (depending on the primer) will anneal the primers to their complementary sequence; and the third one, at 72° C, is the optimal temperature for DNA polymerase to extend the primers.
3. **(5 points) The following enzymes and other molecules are involved in DNA replication “in vivo”. For each enzyme or molecule listed discuss the specific function.**
 - a. DNA polymerase
 - i. DNA Polymerase replicates the strands of DNA in a 5' to 3' direction working from a primer (because it must work from an existing strand of genetic material).
 - b. Helicase
 - i. Helicase separates the strands of DNA at the ORI (origin of replication).
 - c. Topoisomerase
 - i. Topoisomerase prevents the DNA strand from supertwisting by cutting the DNA strand at its ends and then resealing it as necessary.
 - d. Nucleotides
 - i. Each nucleotide has a nitrogenous base and each nitrogenous base is complementary to another nucleotide. This complementarity makes it possible for DNA to replicate itself from just one strand - each strand can serve as a template.
 - e. Primase
 - i. Primase creates a short RNA primer at the 3' end of an unzipped strand of DNA which polymerase can then follow to replicate the whole strand.
4. **(5 points) For each enzyme and molecule discussed above describe how that specific function is accomplished in the PCR.**
 - a. DNA polymerase → DNA polymerase (taq), extracted from heat resistant bacteria so that it does not denature during the thermal cycling process
 - b. Helicase → replaced in the PCR process by the heat cycle, which performs the same unzipping function
 - c. Topoisomerase → replaced in the PCR process by the heat cycle, which performs the snipping function
 - d. Nucleotides → free nucleotides in dehydrated PCR ball.
 - e. Primase → Primase is not needed because we supply the tube with both a forward and reverse primer so primase has no need to identify one.

5. **(2 point) Today we performed the PCR. Why? What is our final goal?**
 - a. We are using PCR to amplify a fragment of the bacterial rpoB gene. We want to see how the rpoB gene's rifampicin resistant region was mutated after being exposed to rifampicin. Overall we are viewing how bacteria can become resistant to antibiotics.
6. **(2 point) For today's experiment what is the template we use for the PCR?**
 - a. The DNA we put in the tube held the bacteria's genome as a template for the PCR.
7. **(1 point) For today's PCR which gene portion did we amplify?**
 - a. We amplified Rif Resistance Regions of the RNAP Beta Subunit gene.